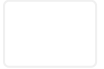


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network shared folder on raspberry pi

 Gary Woodfine / November 27, 2014

One of the really great things about the Raspberry Pi is that it is a Linux based computer, so pretty much anything you can do with a Linux computer you can do with a Pi – within reason!

I have previously written a post about being able to [remotely access the Raspberry Pi via SSH](#)

In this post, I'll provide instructions on how to create a network shared folder on raspberry pi, so you can easily move files back and forth to Pi to a Windows 8 computer or ubuntu desktop.

For this post I will assume you already have connected your Pi via SSH, however you can still following along even if you haven't as all we really need is command line access to the Pi.

Getting set up

To enable network sharing on the Pi we'll need to install a service called Samba.

What is Samba

“Samba is an Open Source/Free Software suite that provides seamless file and print services to SMB/CIFS clients.” Samba is freely available, unlike other SMB/CIFS implementations, and allows for interoperability between Linux/Unix servers and Windows-based clients.

samba.org

Essentially Samba is a little bit of open source goodness that enables linux and windows computers talk to each other.

Install Samba

Installing Samba is fairly trivial task and involves you only running 3 commands in the terminal window on your Raspberry Pi.

SHELL

```
1 sudo apt-get update
2 sudo apt-get install samba
3 sudo apt-get install samba-common-bin
```

After we've installed samba on our pi, we need to carry out 2 configuration steps. Firstly, we need to create a folder we want to share and secondly we'll need to configure samba to actually share the folder.

Create folder to share

Creating a folder to share is really easy, we'll just create a folder and call it **public**.

SHELL

```
1 mkdir public
```

Warning

We are going to create a very permissive share folder on our pi. So if security is a concern, then you may want to read up about the security considerations on Linux

This how-to primarily about creating a share to enable easy access to your Pi over a network

Configure Samba

We now need to edit the samba configuration file, so we'll open the file in nano

SHELL

```
1 sudo nano /etc/samba/smb.conf
```

We need to edit this file in a few locations.

Firstly, because we are going to access the Pi from a Windows 8 computer, we'll need to change

```
SHELL
```

```
1 workgroup = WORKGROUP
```

to

```
SHELL
```

```
1 workgroup = HOME
```

We need to enable WINS support, so Change

```
SHELL
```

```
1 wins support = no
```

to

```
SHELL
```

```
1 wins support = yes
```

Next, we want to enable user security. Remove the hash tag (#) so the entry looks like this:

```
SHELL
```

```
1 curity = user
```

To enhance the network performance, do the same with `TCP_NODELAY`

```
SHELL
```

```
1 socket options = TCP_NODELAY
```

Then we need to create a new share folder information

```
SHELL
1 [public]
2  comment= Pi public share folder
3  path=/home/pi/public
4  browseable=yes
5  writeable=yes
6  valid users = pi
7  guest ok = yes
```

We now need to exit out of nano so Ctrl + X and save it.

We'll run a check on the config file to ensure we haven't broken anything

```
SHELL
1 testparm
```

We should see something like this

```
SHELL
1 Load smb config files from /etc/samba/smb.conf
2 rlimit_max: increasing rlimit_max (1024) to minimum Windows limit (16384)
3 Processing section "[homes]"
4 Processing section "[printers]"
5 Processing section "[print$]"
6 Processing section "[public]"
7 Loaded services file OK.
8 Server role: ROLE_STANDALONE
9 Press enter to see a dump of your service definitions
```

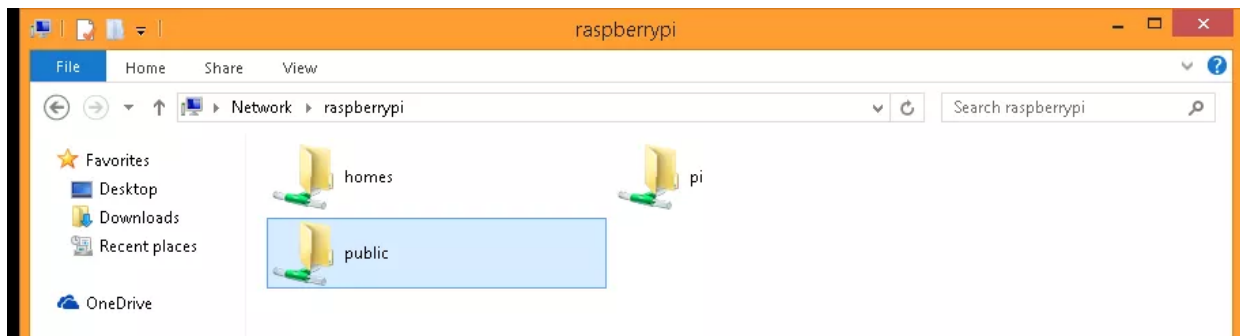
We now need to restart the Samba service in order for it to pick up the new configuration changes

SHELL

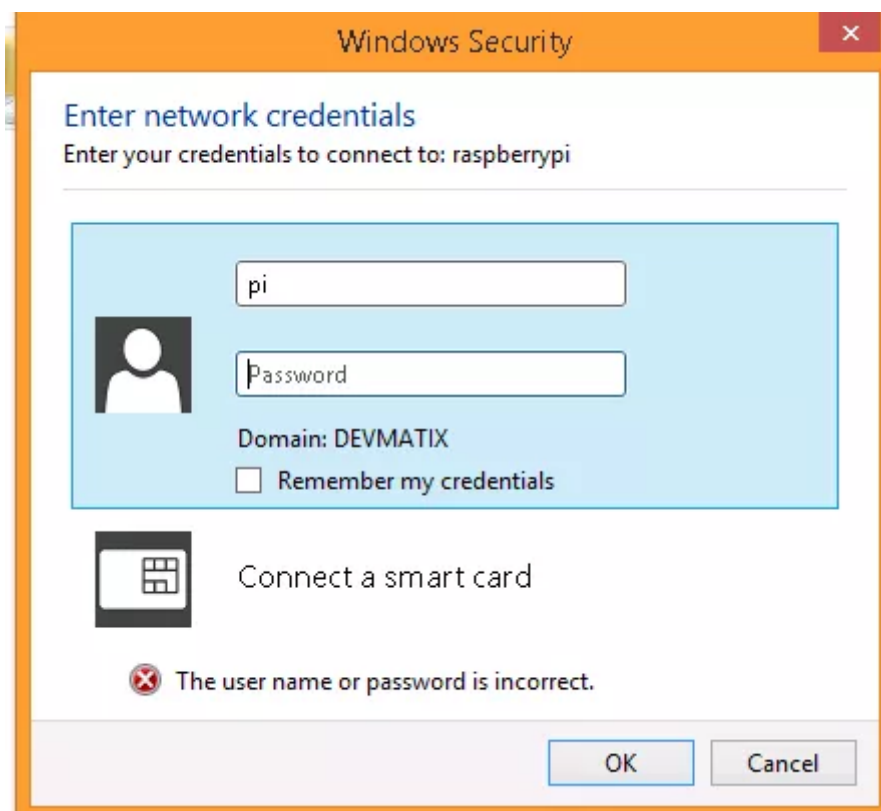
```
1 sudo service samba restart
```

Your Pi should now be sharing files on the network!

You should now be able to access your Raspberry Pi shared folder from you windows machine by open files & folders and navigating to **\\raspberrypi**

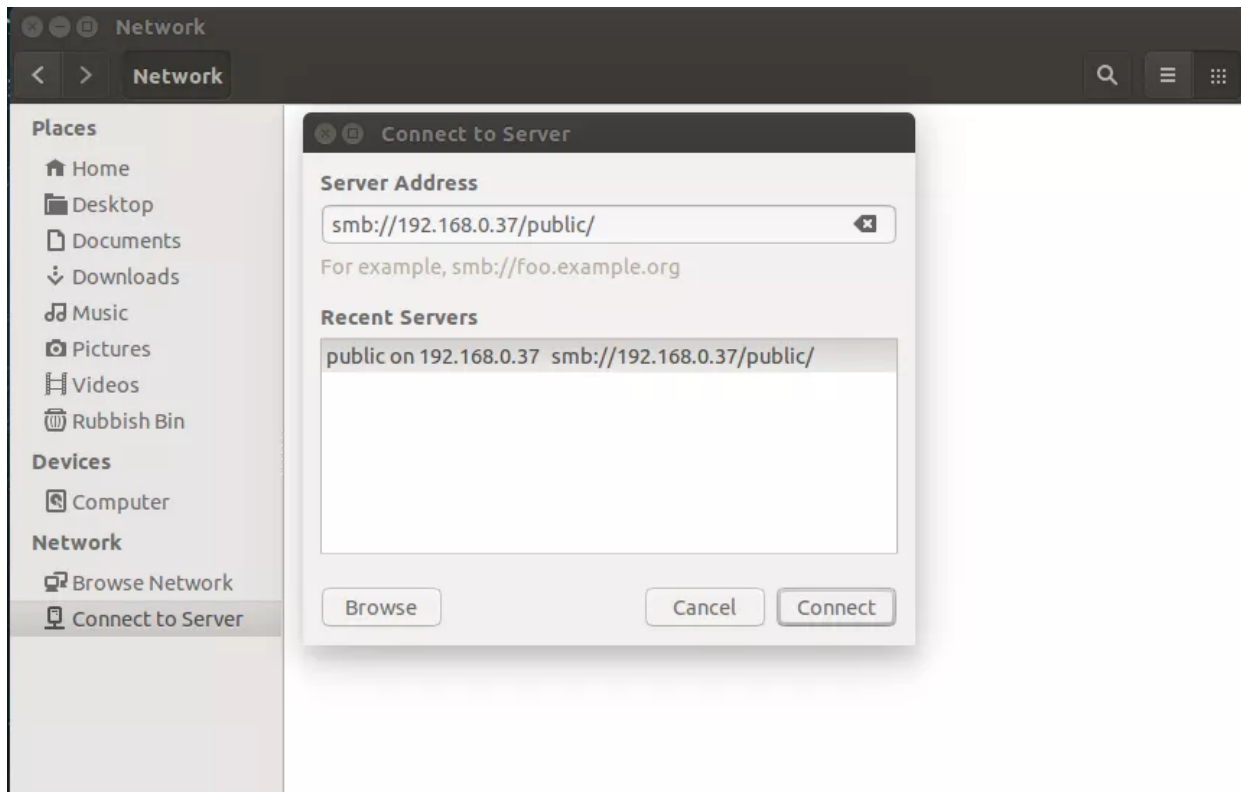


Double clicking on the public folder you may be prompted for a username and password . The default username is **pi** and the default password is **raspberry**



Once you've entered these you are into the shared folder and you can now add, remove and update files as you wish.

To Access the same folder from a Ubuntu machine simply open files and folders and click on **Connect to Server** and enter the same credentials as above, but use the IP address of the Pi



A reminder again that this is a very permissive share and I would strongly advise against leaving it this open, if you plan to connect the Pi to the internet or you're going to store any particularly sensitive information.

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Gary Woodfine

Freelance Full Stack Developer at threenine.co.uk

Helps businesses by improving their technical proficiencies and eliminating waste from the software development pipelines.

A unique background as business owner, marketing, software development and business development ensures that he can offer the optimum business consultancy services across a wide spectrum of business challenges.

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